WEEK 11

Integration and Verification

School data now follows the calendar, the assistant is stricter, the research page is live, and we tested our own result with a pre-registration

Bera Ergin · s29315 · PJAIT · Türksat internship

Four workstreams this week

1 Data and time

The mock was regenerated to follow the calendar; a date gate and RLS hide data after "today".

Date questions now make sense

2 Assistant and SQL

Ungrounded-number check, a correction round and a fallback; two holes in the validator were closed.

Answers rest on queries

3 Research page

An 11-tab report with no login: texts, charts and raw tables from one package.

The system's rule is justified

4 Additional analyses

10 expectations were written and sealed with md5 before the run, then tested in two periods.

5 held · 3 did not · 2 inconclusive

The mock data was regenerated to follow the calendar

Why we lifted the freeze

- In the old mock, grades fell before the school year started, so date questions meant nothing in the demo.
- The generator fixed some conditions by student name. Profile parameters replaced this; an AST test forbids branching on names.
- The demo stories (flagged and unflagged students) are protected by a generation gate.

36
students · 6 classes

1,440
grade records

6,444
attendance rows

179
school days

```
seed 20260914 · 14.09.2026–25.06.2027
md5 de7da13d07102011882382e467968e24
```

Date gate: the system does not see past "today"

Component	What it does
`takvim.py`	MEB 2026–2027: two terms, mid-term breaks, semester break, public holidays
`DEMO_TODAY`	The demo's "today"; if empty, the Europe/Istanbul date
`app.today + RLS`	Restrictive policies on grades and attendance
`student_rank`	Date condition; empty values sorted last
`Risk window`	Risk is computed from the current term's window

A leak was closed

students.total_absences carried the year-end total: in October, June's absences were visible, and it disagreed with the attendance table.

The column was removed. Absences now come from attendance everywhere, with excused and unexcused counted separately.

Assistant: every number must rest on a query

Date block

Each request gives the model today, the term, the visible data range and a calendar summary.

Context

Number check

Derived numbers are computed in SQL. If an ungrounded number appears, the rejected numbers are listed and one extra tool call is allowed (4 at most).

No invented numbers

Fallback

If the correction round also fails, the query results are shown; the event is written as a structured log.

No silent failure

Fixed: numbers from earlier turns are now allowed; month name + number was masked wrongly. Known limit: the abs() match does not check direction.

Two holes in the SQL validator were closed

Problem	Effect	Fix
AND, CASE, NULLIF were treated as functions	Valid queries were rejected; the assistant said "I couldn't reach the records"	Operators separated from the function check; NULLIF allowed
DELETE could pass inside a CTE	The read-only guarantee could be broken	DML, DDL, INTO and locks rejected everywhere, including CTEs and subqueries

The validator allows the public schema only. New test: every example SQL in the assistant's instructions passes the validator.

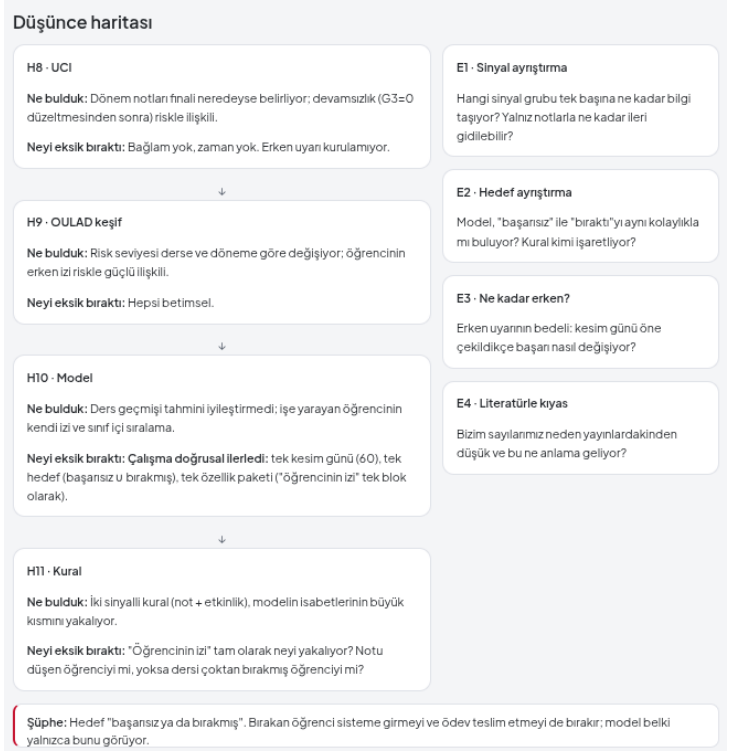
Research page: the report behind the rule



Summary tab of the Turkish interface. Screenshot taken before the 20% format fix.

- No login; it does not touch school data or the backend.
- 11 tabs, each with an "In this section" box.
- Texts, charts and raw tables (CSV) from one package: 52 images.
- No hand-typed numbers; placeholders in the text are resolved from the package.
- No findings are drawn from the synthetic school data.

Turning point: we doubted our own result



> *The target is "failed or withdrew". A student who withdraws also stops logging in and submitting; perhaps the model sees only that.*

If so, the risk tab shows not the struggling student but the one who has already left.

Four questions

E1 · Which signal carries the information?
E2 · Who is easier to find?
E3 · How early?
E4 · What does the literature say?

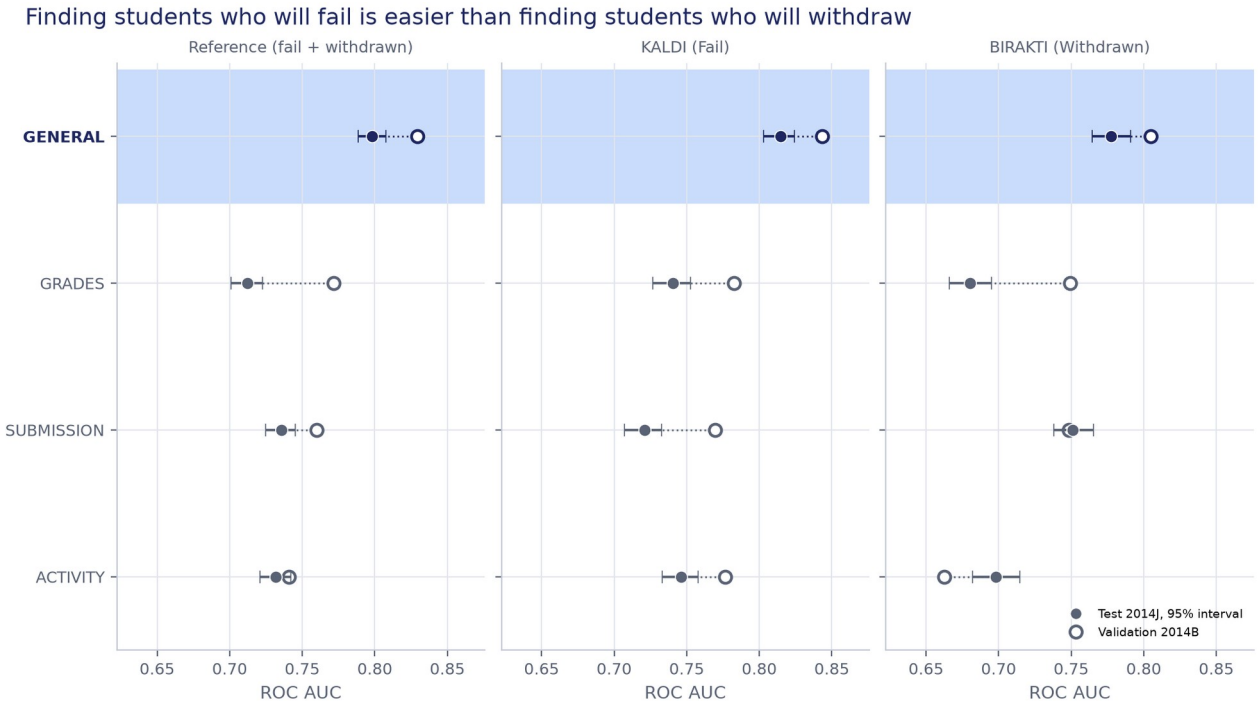
Written first, sealed, then run

10 expectations written before the run

B1 No single group reaches GENERAL ?	**B2** GRADES is the weakest single group ?	**B3** Model without activity stays close ?	**B4** BIRAKTI is easier than KALDI ?	**B5** ACTIVITY is strong for BIRAKTI ?
B6 GRADES share grows for KALDI ?	**B7** Rule hits are mostly withdrawals ?	**B8** Performance rises with cutoff day ?	**B9** Most of the gain comes early ?	**B10** ACTIVITY carries day 14 ?

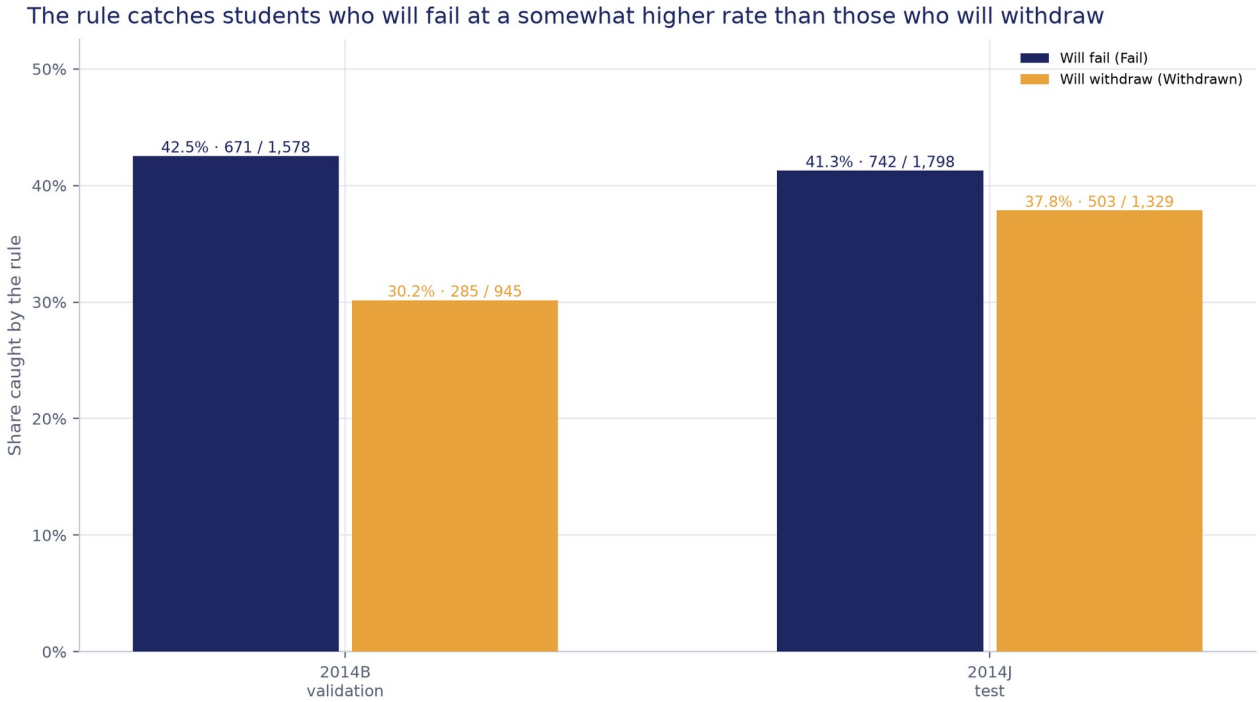
10 expectations were written with numeric criteria and the files' md5 was recorded; the analysis code refuses to run if the md5 does not match. Rule: held in both periods = held, in neither = did not hold, split = inconclusive.

The doubt failed: future failures are easier to find



- KALDI (Fail): ROC AUC 0.843 validation, 0.815 test.
- BIRAKTI (Withdrawn): 0.805 and 0.778.
- We expected the opposite: B4 did not hold.
- For BIRAKTI the strongest single group is submission (0.751), not activity: B5 did not hold.

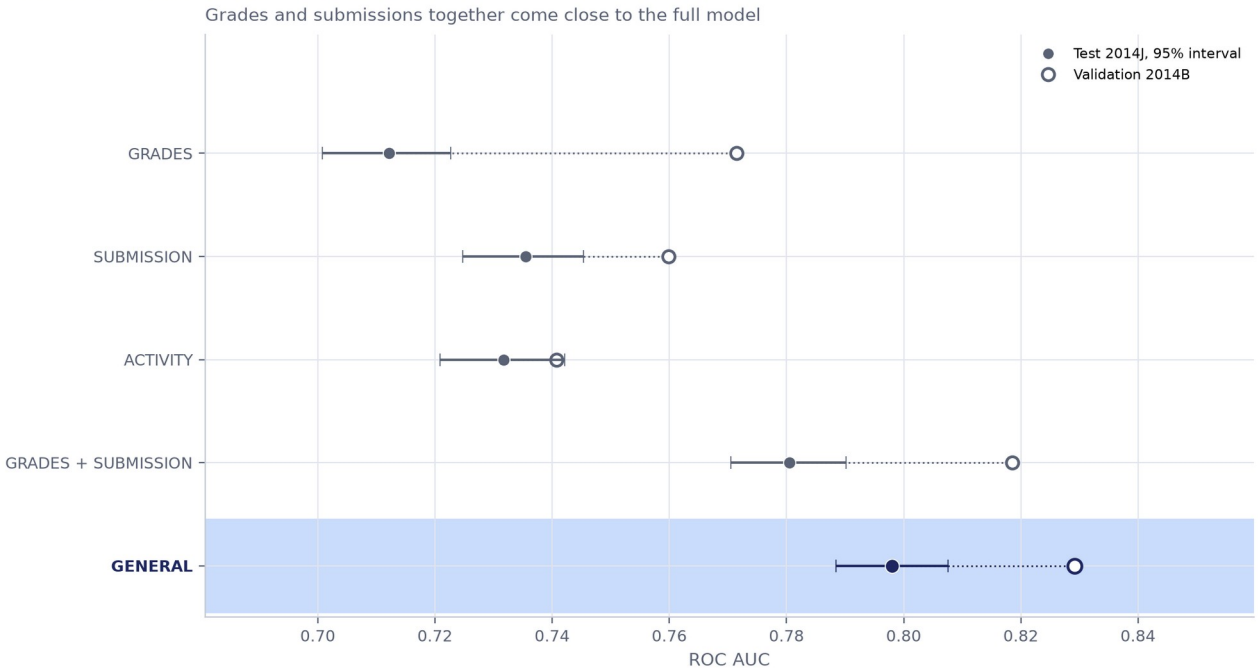
Whom does the rule catch?



- 42.5% (validation) and 41.3% (test) of future failures were flagged.
- 30.2% and 37.8% of future withdrawals.
- In test, 742 hits are Fail and 503 Withdrawn: most do not come from withdrawals (B7 did not hold).
- The risk tab also shows struggling students.

Where is the information? Grades and submission together

A single signal group is not enough



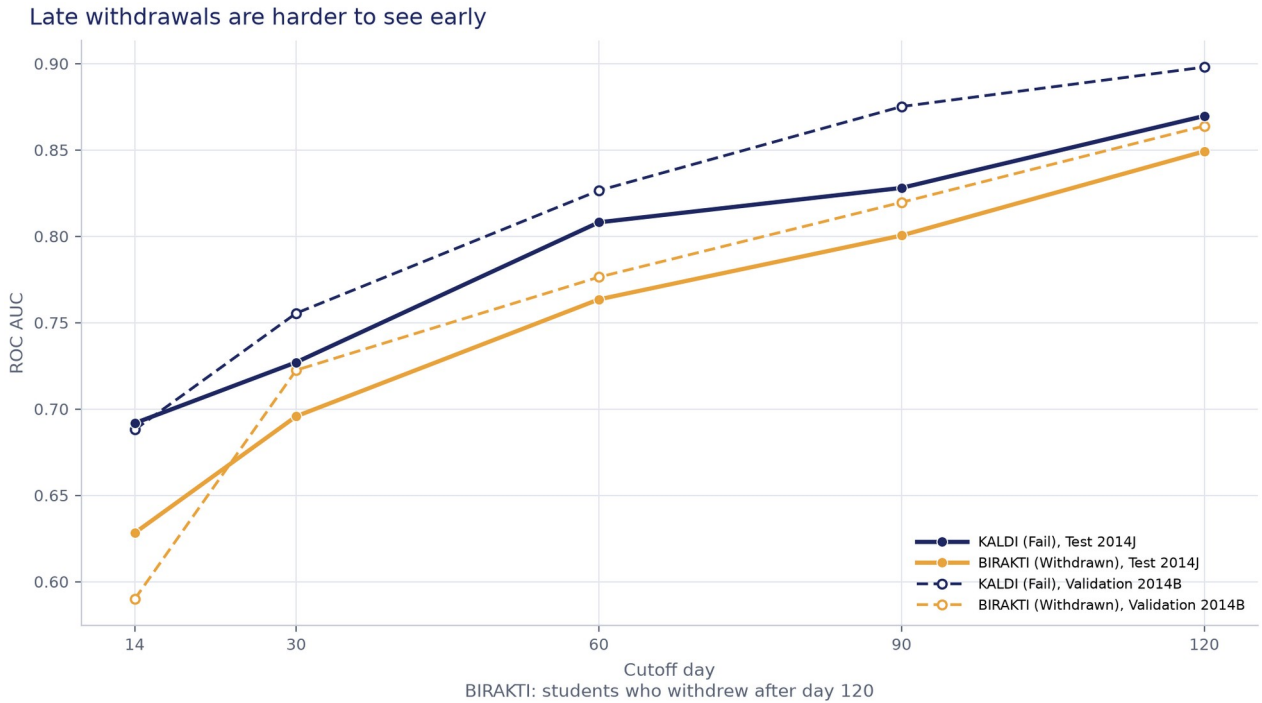
- No single group reaches the full model (B1 held).
- Grades + submission trail the full model by 0.011 (validation) and 0.017 (test) (B3 held).
- The strongest single group changes between periods (B2 inconclusive).
- Most valuable missing signal for the school: assignment submission records.

How early? Most of the gain comes by day 60



- Fixed cohort, test: day 14 0.675 · day 30 0.711 · day 60 0.788 · day 120 0.864.
- Performance rose at every step (B8); most of the gain came early (B9).
- On day 14, activity is almost as good as the full model: 0.669 vs 0.675 (B10).
- Day 60 is justified in hindsight.

Late withdrawals are harder to see early



- On day 14: KALDI 0.692, BIRAKTI 0.628.
- The gap narrows by day 120: 0.870 and 0.849.
- BIRAKTI here means withdrawing after day 120; early withdrawals are not in the cohort.
- Early warning sees withdrawal risk only weakly.

Literature: the closest design gives similar numbers

Study	Prediction point	Demographics	Reported value
This study	Day 30, fixed cohort	No	ROC AUC 0.711
Wu, Jia and Pham (2026)	Day 28, temporal	No	ROC AUC 0.709
Wu et al. (2026)	Day 28, grouped	Yes	ROC AUC 0.775
Adnan et al. (2021)	20% of course	Yes	Precision 0.79

Wu et al.: Frontiers in Education 11 · Adnan et al.: IEEE Access 9. Split type not verified for Adnan et al. and Scientific Reports (2025).

- Published values above 90% come from other prediction points, splits and targets; they are not directly comparable.
- In Wu et al., withdrawal is also harder to predict (0.697 vs 0.720): our B4 result replicates independently.
- This is not a validation; it signals our numbers are in a plausible range for this protocol family.

Result: 5 held, 3 did not, 2 inconclusive

10 pre-registered expectations: 5 held, 3 did not hold, 2 inconclusive

B1	B2	B3	B4	B5
No single group reaches GENERAL	GRADES is the weakest single group	Model without activity stays close	BIRAKTI is easier than KALDI	ACTIVITY is strong for BIRAKTI
Held	**Inconclusive**	**Held**	**Did not hold**	**Did not hold**

B6	B7	B8	B9	B10
GRADES share grows for KALDI	Rule hits are mostly withdrawals	Performance rises with cutoff day	Most of the gain comes early	ACTIVITY carries day 14
Inconclusive	**Did not hold**	**Held**	**Held**	**Held**

The three that did not hold (B4, B5, B7) were the three legs of the doubt that "the model only finds withdrawals". B11 (literature) is supported; the sample is small.

What changed, what did not

Changed

- The doubt failed; we know what the system catches.
- The day-60 choice is justified.
- The most valuable missing signal for the school is identified: submission records.

Unchanged

- The rule in the system.
- The analyses were for understanding; any change would be tested separately under a new pre-registration.

Teacher effect: not measurable, neither dataset has teacher data. The system is built to be insensitive to it (within-class relative rank); it must not become a teacher-ranking tool.

Bug log 1: data and assistant

Bug	Root cause	Fix
Grades before school started	The mock was generated without the calendar	Deterministic regeneration on the calendar
Generator fixed by name	Conditions were met with special code for one student	Profile parameters; no-branching-on-names test
total_absences leak	The year-end total was visible on every date	Column removed; absences from attendance
"Couldn't reach the records" dead end	The validator treated AND, CASE, NULLIF as functions	Operator set separated
DELETE inside a CTE	The check did not cover CTE bodies	DML, DDL, INTO, locks rejected everywhere
Assistant numbers	Earlier-turn numbers not allowed; month name + number masked wrongly	Added to the allowed set; masking fixed

Bug log 2: analysis and tests

Bug	Root cause	Fix
Fragile UI tests	Cards were counted by their border	Tests find them by their own markers
In-cell computation in notebooks	Computations in 01, 02, 03 lived in the notebook	Moved to the package; outputs identical
Wrong Jupyter kernel	The copied .venv pointed to week08	staj-week11 kernel
Pre-registration check	It read the first md5 record	It reads the latest record
feature_list groups	Week 9 "school counterpart" wrong; vle_click_* outside groups	Fixed; prefix rule and test
Structure test broke	After the text patch, placeholders were searched unresolved	Test bound to the page's resolver

> *Open: silent method switch via except Exception in explain.py (did not affect today's output) · UCI TR images are in English (Week 12).*

Verification

1,210 passed

3 skipped · full pytest, 7 November

```
python -m data_science.oulad.ek
python -m data_science.oulad.ek --e3
python -m data_science.oulad.figures --rule2 --ek
python scripts/arastirma_paketi.py
python -m pytest -q
```

- Package diff: no existing key changed value; the only new key is rule.flag_rate (from the rule's own constant).
- Pre-registration: three files sealed with md5; ek.py does not run if the latest record does not match.
- Parquet was rewritten: 26,403 rows, same splits; e1, e2, e3 and expectations IDENTICAL.
- Notebook 01–03 computations moved to the package; outputs identical.

Known limits and Week 12

Known limits

- TR versions of the UCI and OULAD exploration charts are in English or use decimal points; plotting will move to the package.
- explain.py: the silent method switch will be fixed.
- Group assignment lives in two places (dataset.py, ek.py).
- Assistant: the abs() match does not check direction.

Week 12

- get_risk tool: for teachers and principals; parents and students get "talk to the class teacher".
- Security before deploy: SECRET_KEY, login counter, question length, log masking.
- Docker on the Contabo VPS; notebooks rerun and compared in a clean environment.
- README, KVKK scope, demo.